

Table 1

CMS solutions: A quick comparison			
Description	Alfresco based CMS	Drupal based CMS	Joomla! based CMS
Application server	Jboss, Apache	Apache, IIS, or any PHP supported application server	Any Java supported application server
Operating system	Windows, Linux, Solaris	Windows, Linux, Solaris	Any
Language (for Web design)	Java (JSF, JavaScript, Ajax)	PHP	PHP, JavaScript, Ajax support
Database (RDBMS)	Oracle, SQLServer, etc	MySQL, PostgreSQL	MySQL
DMS (Document management system)	Inbuilt	Can use Alfresco for extended facilities	Inbuilt
Multilingual support	Supports internationalisation	Supports internationalisation	No
End user accessibility	Average speed	High speed	Below average speed
Server (machine) requirement	Requires really high-end configuration	Can manage easily with average configuration	Requires really high-end configuration
Other facilities	Portal support, jBPM support for workflow management	RSS feed, blog	Page caching, RSS feed, blog, search

of the flexibility of these tools when combined with our CMS solution. For example, if we need support for PDF printing or barcode printing, we need to check the feasibility of offering these features with tools like Adobe Indesign or iText.

We need to analyse our competitors and design a solution that not only offers all the facilities provided by them, but even more. Also, the designer has to analyse the support for a multilingual system or for internationalisation (i18n) if required for the system, so that it caters to users across various countries. i18n support is not an easy-to-handle design as it requires deep study into the type of language support (like Unicode support) and locale-specific data requirements (like country-wise time zones, currencies, etc).

Also, the designer has to keep in mind other technological and logistic requirements like the performance of the system, its reliability (it should not hang, restart or respond inappropriately to the user), development related roadblocks like insufficient time for development, etc. It should also be easy/flexible to maintain, modify and deploy, and designers should factor in changing resource (systems and hardware) needs and network traffic.

Deciding and planning for these requirements will make the CMS solution comprehensive and robust for users, while being easy and flexible to implement, handle and maintain for the developers.

Drupal

Drupal is a growing CMS solution which uses PHP for Web designing (the front-end). This is a flexible and lightweight system, which can sit on application servers like Apache or Microsoft IIS.

We can configure Drupal easily with a database system based on MySQL or PostgreSQL. Drupal has several

advantages like i10n support (internationalisation), a faster solution and ACL (access control list) provisioning (user handling with access permissions), to name a few.

The drawbacks include poor support (community users are few) and less support for document management (DM). Recently, Drupal developed a package for Alfresco integration to support better document management.

Joomla!

Joomla! is a popular open source CMS which is flexible and has a wide variety of content themes (Web design models or skeletons of Web page designs). It has good support for blogging and DM, and from different communities and forums. The drawbacks are that it has support only for Apache and has a poor ACL (user management). Also, being a heavyweight results in slower and poor performance.

Alfresco

Alfresco is the leading CMS solution based on Java and has very good support on a wide variety of linked applications/tools and facilities. It has the support of a large development community (blogs and forums) which uses it across different domains.

It is not so lightweight and not as fast as Drupal. Another notable drawback is that setup and configuration are comparatively tedious.

Table 1 gives a comparison of these CMS solutions. 

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